

INDIAN INSTITUTE OF TECHNOLOGY, ROPAR

TINKERING PROJECT



Maze-Solving Robot using Servo and Ultrasonic Sensors

GROUP NO.: 04

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Project Overview

The Maze-Solving Robot is designed to navigate through unknown environments autonomously using ultrasonic sensors for obstacle detection and servos for controlled movement. This project focuses on real-time path planning to solve mazes efficiently without requiring external guidance. The robot can be utilized in various applications such as autonomous navigation, rescue missions, and industrial automation.

Objective

The primary goal of this project is to develop an intelligent robot that can analyze its surroundings, detect obstacles, and navigate through complex mazes using optimized algorithms. By integrating ultrasonic sensors and servo motors, the robot can dynamically adjust its path to reach the desired destination.

Components and Their Functions

1. Arduino (Microcontroller) (1 pcs)

- Acts as the brain of the robot.
- Processes sensor data and executes navigation algorithms.



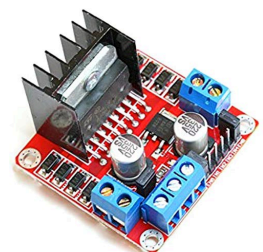
2. Ultrasonic Sensor (HC-SR04) (3 pcs)

- Used for detecting obstacles by measuring distance.
- Helps in real-time mapping of the maze.



3. H-Bridge Motor Driver (L298N) (1 pcs)

- Controls the speed and direction of the motors.
- Acts as an interface between the microcontroller and motors



4. **Chassis(plastic sheet) and Wheels (1 set/ 4 wheels)**

- Provide the physical structure of the robot.
- Enable smooth movement across different surfaces.



5. **9V Rechargeable Battery / Holder (2 pcs)**

- Supplies necessary voltage and current for the robot to function.
- Ensures uninterrupted operation during navigation.



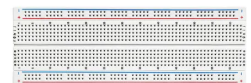
6. **BO Motor (2 pcs)**

- Provide the necessary drive for the robot's wheels.
- Enable smooth and controlled movement within the maze.



7. **Breadboard (1 pcs)**

- Facilitate easy connections between different components.
- Enable prototyping without soldering.

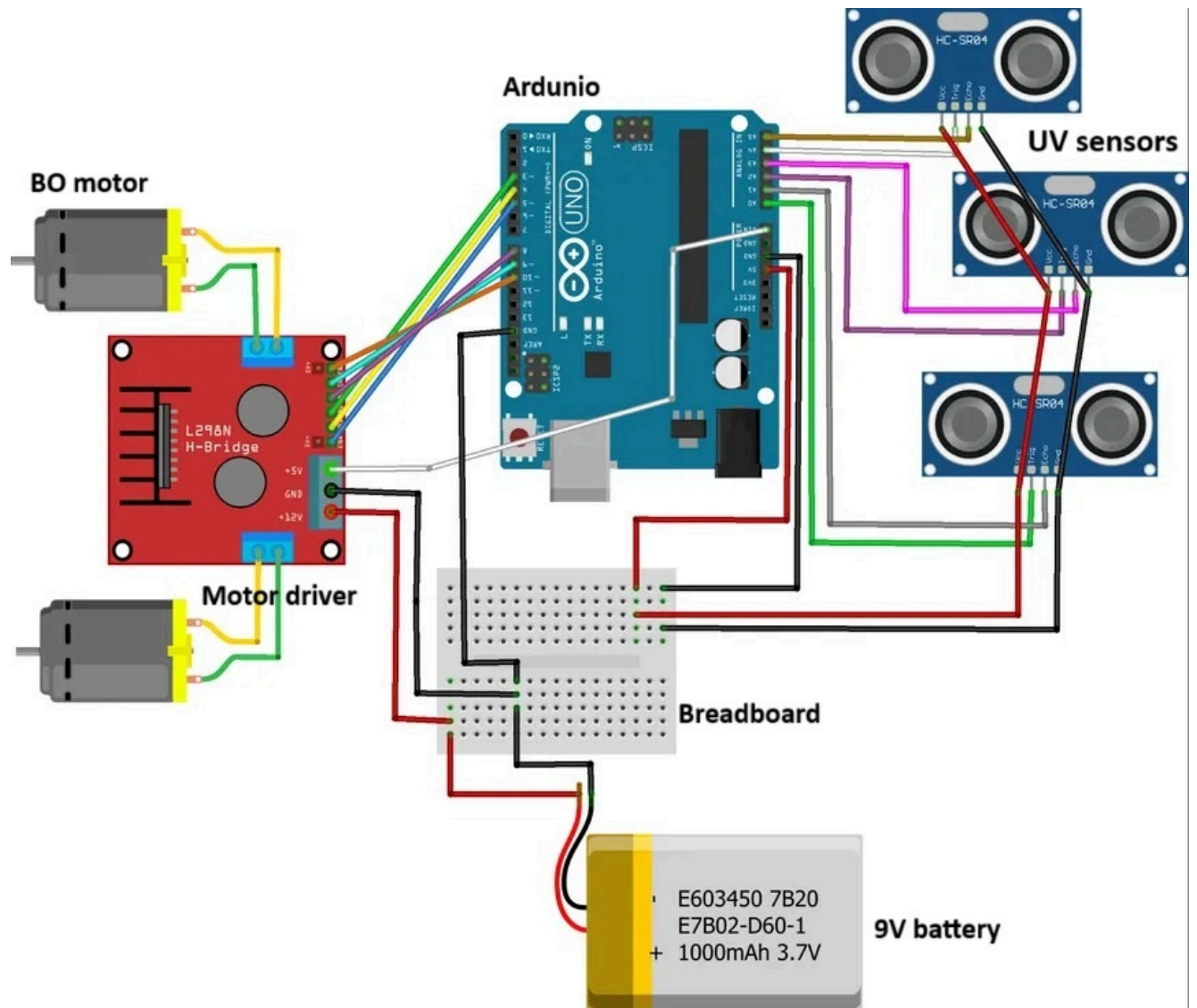


8. **Wires(3 set)**

- Connect various electronic components.
- Ensure proper electrical communication between modules.



Circuit Diagram:



Challenges faced by us :

1. As the robot goes in the wrong path where it gets stuck, as it sees a wall in front, left and right. At that time it won't be able to go anywhere and just move forward and backward.

Solution:

Solution to this problem is that the robot detects this problem and moves backward slowly as it did not turn 180 degrees around but it moves by some angle and comes on the right path to get out of the maze.

2. The second challenge faced by us is at the timing of turning to the right or left. It did not turn precisely as sometimes it moves very early or sometimes it turns very late.

Solution:

According to our code the turning is done like one wheel moves in forward direction while the other moves in reverse direction. We use a hit and trial method to set the accurate speed of each wheel to take a precise turn at the time of turning to the right or left.